

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Semester Test 1

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Module EBN122

Elektrisiteit en Elektronika

25 Augustus 2016

Module EBN122

Electricity and Electronics

25 August 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute 90 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	15 + Answer sheet		

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. J. Joubert
	3. R dos Santos

VRAAG/QUESTION 1 [8]

Question 1.1 [2]

The kitchen light in a household is used for 8 hours every day per year (365 days). If the lightbulb was a 100 W candescent lightbulb, what is the savings over a one year period if the lightbulb is replaced with a 20 W LED light? Assume electricity costs R1.50 per kWh.

Vraag 1.1 [2]

'n Gloeilamp in 'n kombuis word vir 8 ure elke dag van die jaar (365 dae) gebruik. Indien die gloeilamp met 'n gradering van 100 W vervang word met 'n LED lamp van 20 W, wat is die besparing oor 'n een jaar periode? Aanvaar elektrisiteit kos R1.50 per kWh.

$$\text{Bulb 1: } \frac{100 \text{ W} \times 8 \text{ hours}}{1000} \times 365 \text{ days} \times R1.50 / \text{kWh} = R438$$

$$\text{Bulb 2: } \frac{20 \text{ W} \times 8 \text{ hours}}{1000} \times 365 \text{ days} \times R1.50 / \text{kWh} = R87.6$$

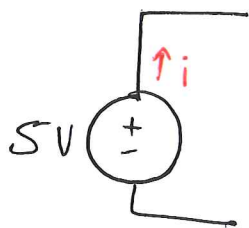
$$\text{Savings: } R438 - R87.6 = \underline{R350.4} \checkmark$$

Question 1.2 [2]

A 5V battery is connected in series with an element. The current exiting the positive terminal of the battery decreases linearly from 20 mA to 5 mA between $t = 0$ s and $t = 20$ s, while the voltage remains constant. What is the energy associated with the battery?

Vraag 1.2 [2]

'n 5 V battery word in serie met 'n element gekoppel. Die stroom wat by die positiewe terminal van die battery uitvloei val liniêr van 20mA tot 5 mA tussen $t = 0$ s en $t = 20$ s. Wat is die energie geassosieer met die battery?



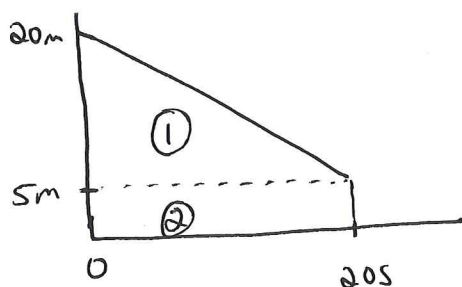
$$W = \int P dt$$

$$= \int -V I dt$$

$$\cancel{W} = -5 \int I dt$$

$$= -5 \times 250 \times 10^{-3}$$

$$= \underline{-1.25 \text{ J}} \checkmark$$



$$\text{Area 1: } \frac{1}{2} \text{ base} \times \text{height}$$

$$= \frac{1}{2} (20) \times 15 \times 10^{-3}$$

$$= 150 \times 10^{-3}$$

$$\text{Area 2: } \text{base} \times \text{height}$$

$$= 5 \times 10^{-3} \times 20$$

$$= 100 \times 10^{-3}$$

$$\int I dt = \text{Area 1} + \text{Area 2} = 100 \times 10^{-3} + 150 \times 10^{-3}$$

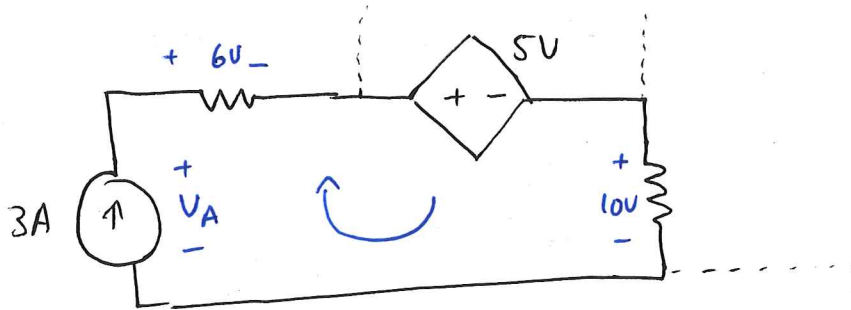
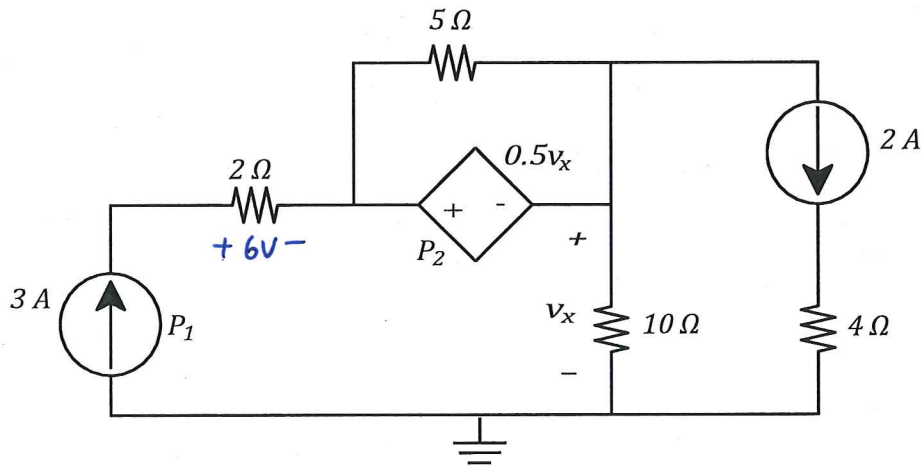
$$\int I dt = 250 \times 10^{-3}$$

Question 1.3 [4]

Determine P_1 and P_2 if $V_X = 10$ V.

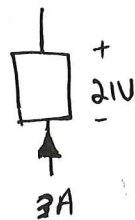
Vraag 1.3 [4]

Bepaal P_1 en P_2 indien $V_X = 10$ V.

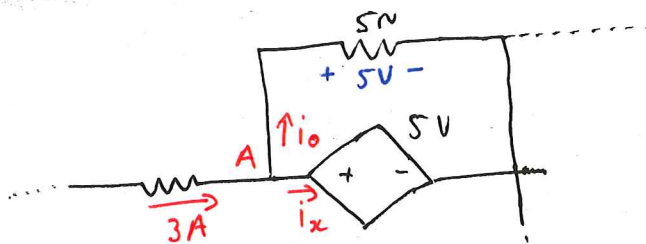


$$\text{KVL: } -V_A + 6 + 5 + 10 = 0$$

$$V_A = 21\text{V}$$



$$\begin{aligned} P_1 &= -VI \\ &= -21 \times 3 \\ &= -63\text{W} \end{aligned}$$



KCL at node A: $3 = i_x + i_o$

$$\text{but } i_o = \frac{5}{5} = 1\text{A}$$

$$\therefore i_x = 2\text{A}$$



$$\begin{aligned} P_2 &= +VI \\ &= 2 \times 5 \\ &= 10\text{W} \end{aligned}$$

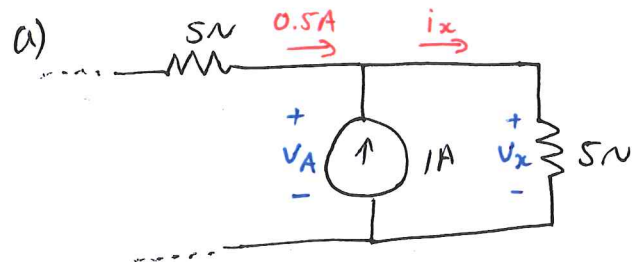
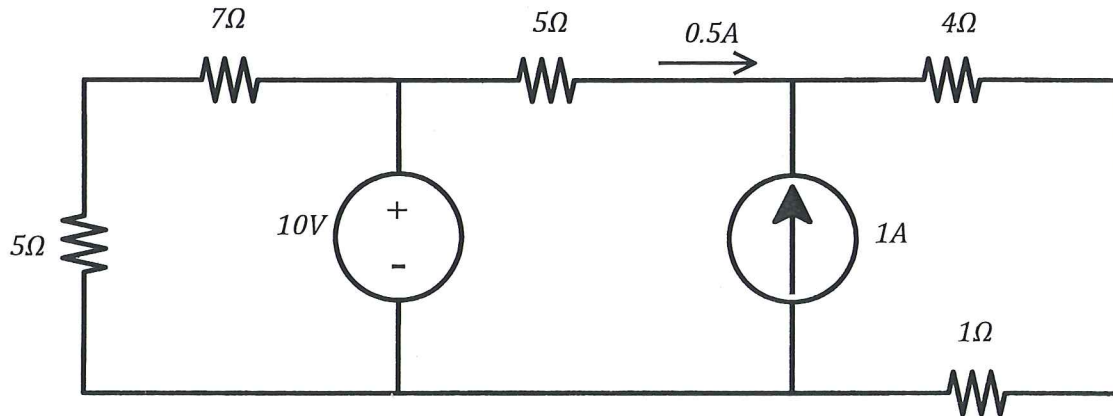
VRAAG/QUESTION 2 [16]

Question 2.1 [4]

- a) Determine the power associated with the current source.
b) Determine the power associated with the 7Ω resistor.

Vraag 2.1 [4]

- a) Bepaal die drywing ge-assosieer met die stroombron.
b) Bepaal die drywing ge-assosieer met die 7Ω weerstand.

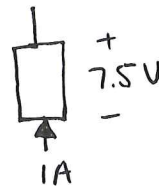


$$\text{KCL: } 0.5 + 1 = i_x$$

$$i_x = 1.5 \text{ A}$$

$$V_x = 1.5 \times 5 = 7.5 \text{ V}$$

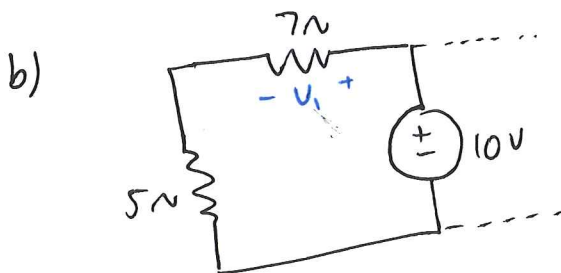
$$V_A = V_x \text{ (parallel)}$$



$$P_A = -VI$$

$$= -7.5 \times 1$$

$$= \underline{-7.5 \text{ W}}$$



$$V_1 = \frac{7}{7+5} (10) \text{ (Voltage division)}$$

$$V_1 = 5.83 \text{ V}$$

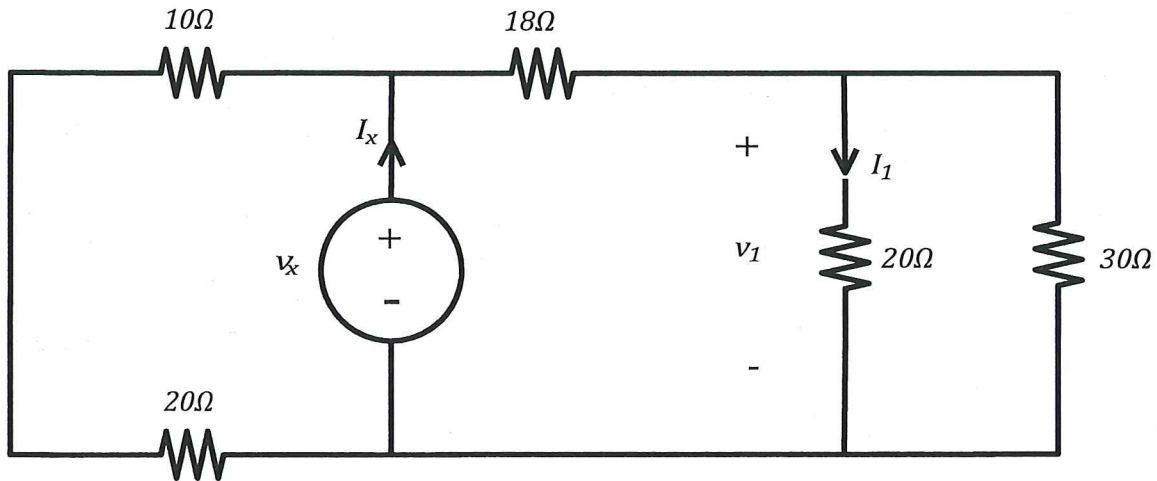
$$P = \frac{V^2}{R} = \frac{(5.83)^2}{7} = \underline{4.86 \text{ W}}$$

Question 2.2 [4]

- a) Determine v_1 if $v_x = 15\text{V}$.
b) Determine I_1 if $I_x = 3\text{A}$.

Vraag 2.2 [4]

- a) Bepaal v_1 as $v_x = 15\text{V}$.
b) Bepaal I_1 as $I_x = 3\text{A}$.



a) $20 // 30 = 12\Omega$

$$v_1 = \frac{12 \times 15}{18 + 12} = 6\text{V} \checkmark \checkmark$$

b)

$$i_A = \frac{30 (3)}{30 + 30} = 1.5\text{A}$$

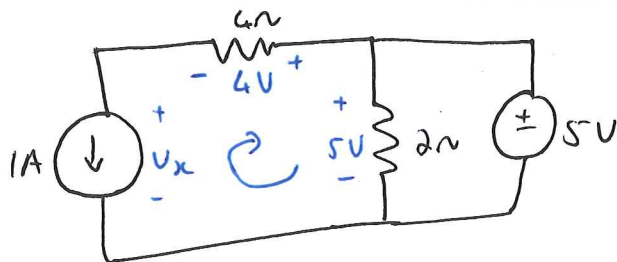
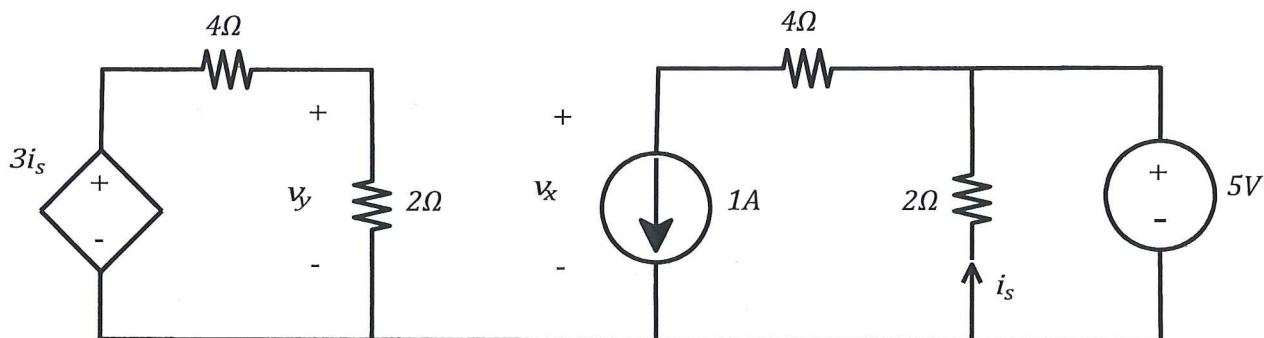
$$i_1 = \frac{30 (1.5)}{20 + 30} = 0.9\text{A} \checkmark \checkmark$$

Question 2.3 [4]

Determine V_x and V_y .

Vraag 2.3 [4]

Bepaal V_x en V_y .

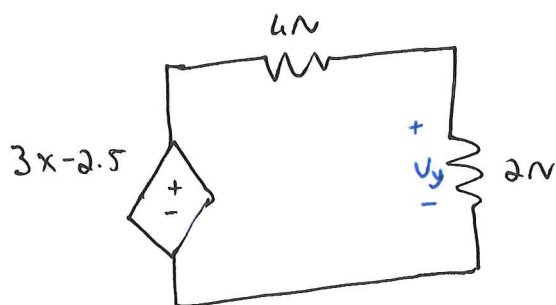


$$\text{KVL: } -V_x - 4 + 5 = 0$$

$$\underline{1V = V_x} \quad \checkmark \checkmark$$

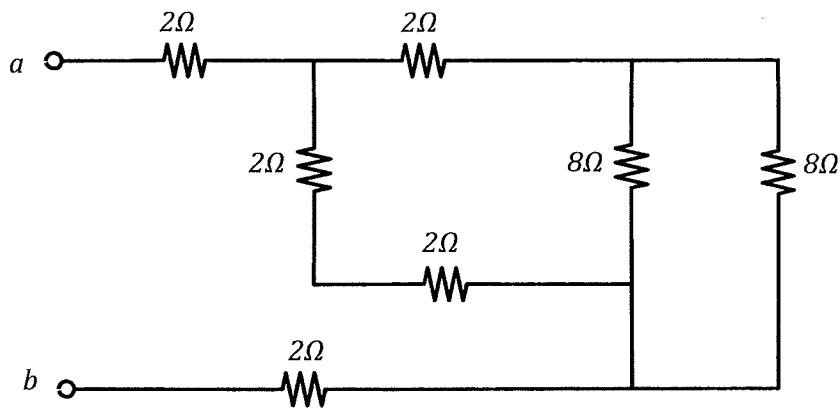


$$i_s = -\frac{5}{2} = -2.5 \text{ A}$$

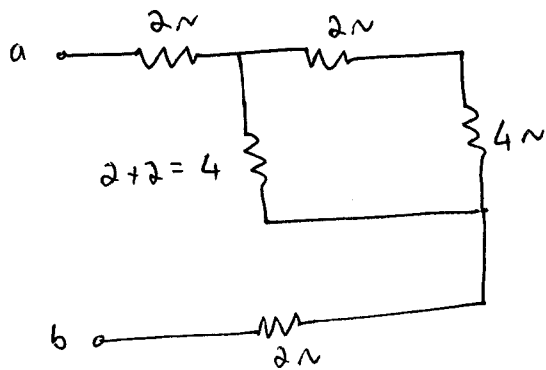


$$V_y = \frac{2}{2+4} (3x-2.5) \quad (\text{Voltage division})$$

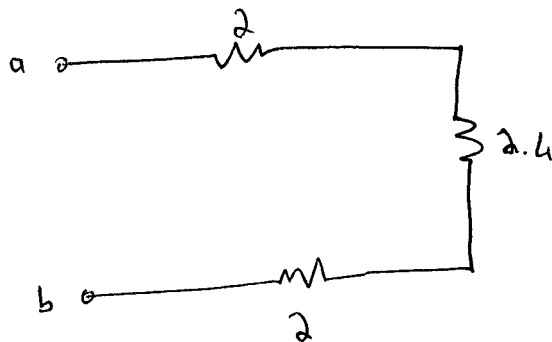
$$\underline{V_y = -2.5 \text{ V}} \quad \checkmark \checkmark$$

Question 2.4 [2]Find the equivalent resistance wrt terminals $a-b$.**Vraag 2.4 [2]**Bepaal die ekwivalente weerstand tov terminale $a-b$.

$$8 // 8 = 4\Omega$$



$$(2 + 4) // 4 = 2.4\Omega$$



$$R_{ab} = 2 + 2 + 2.4$$

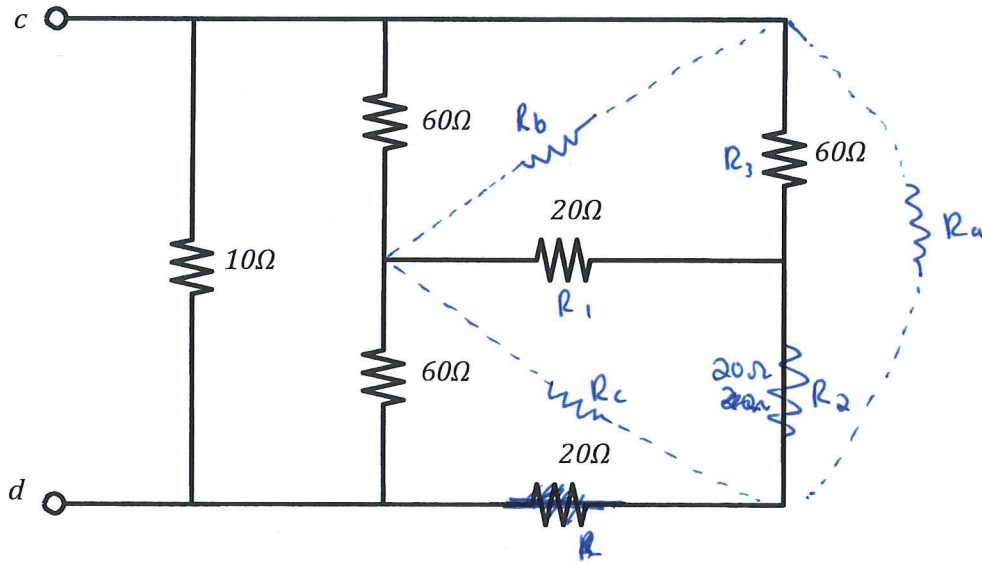
$$= 6.4\Omega \quad \checkmark \checkmark$$

Question 2.5 [2]

Find the equivalent resistance wrt terminals c-d.

Vraag 2.5 [2]

Bepaal die ekwivalente weerstand tov terminale c-d.



$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = \frac{20 \cdot 20 + 20 \cdot 60 + 60 \cdot 20}{20} = \frac{2800}{20}$$

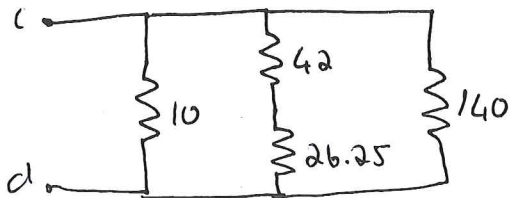
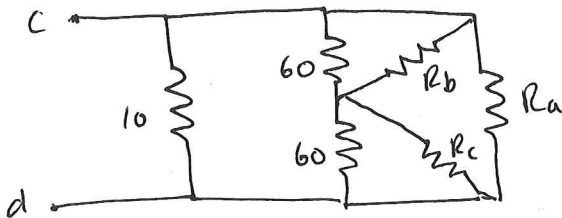
$$R_a = 140 \Omega$$

$$R_b = \frac{2800}{20} = 140 \Omega$$

$$R_c = \frac{2800}{60} = 46.67 \Omega$$

$$60 // R_b = 60 // 140 = 42 \Omega$$

$$60 // R_c = 60 // 46.67 = 26.25 \Omega$$



$$R_{cd} = \left[\frac{1}{10} + \frac{1}{42 + 26.25} + \frac{1}{140} \right]^{-1} = 8.21 \Omega$$

QUESTION/VRAAG 3 [22]

Question 3.1 [4]

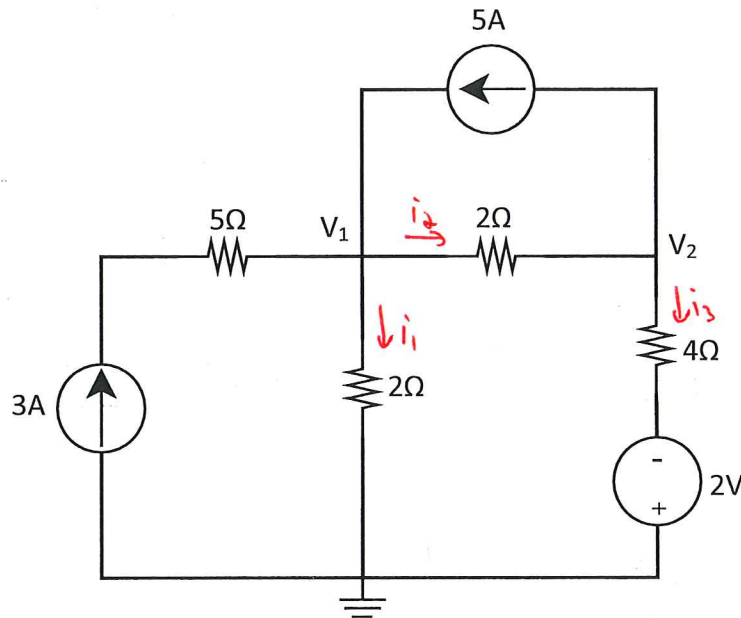
Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} 2V_1 + aV_2 &= b \\ 2V_1 + cV_2 &= d \end{aligned}$$

Vraag 3.1 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} 2V_1 + aV_2 &= b \\ 2V_1 + cV_2 &= d \end{aligned}$$



At node V_1 : $3 + 5 = i_1 + i_2$

$$8 = \frac{V_1}{2} + \frac{V_1 - V_2}{2}$$

$$16 = V_1 + V_1 - V_2$$

$$16 = 2V_1 - V_2$$

$$a = -1 \checkmark \quad b = 16 \checkmark$$

At node V_2 : $i_2 = 5 + i_3$

$$\frac{V_1 - V_2}{2} = 5 + \frac{V_2 - (-2)}{4}$$

$$2V_1 - 2V_2 = 20 + V_2 + 2$$

$$2V_1 - 3V_2 = 22$$

$$c = -3 \checkmark \quad d = 22 \checkmark$$

$$a = -1 \quad b = +16 \quad c = -3 \quad d = 22$$

$$\text{or } a = -3 \quad b = 22 \quad c = -1 \quad d = 16$$

The equations given in the question were not clear as to which node each equation represented. Therefore, the value of $a = -1$ or -3 , $b = 16$ or 22 , etc.

However, if you received a mark for $a = -1$, and you wrote $c = -1$, you will only receive a mark for $a = -1$ and not c .

Question 3.2 [4]

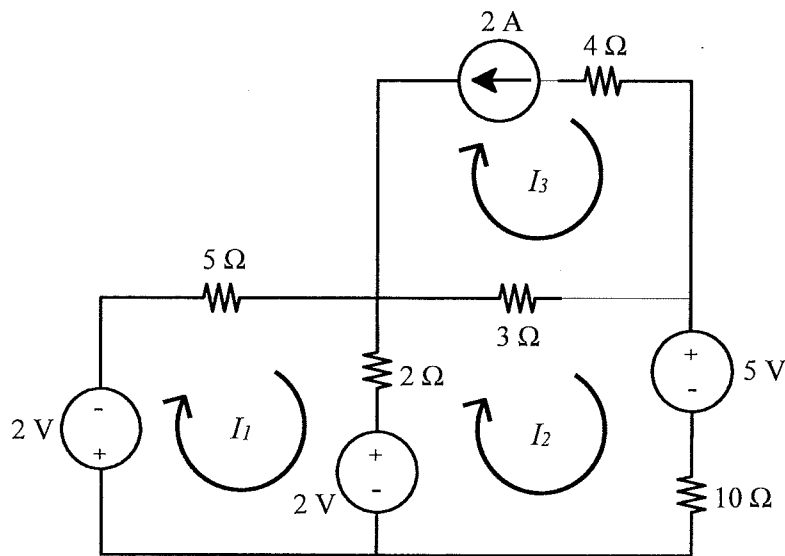
Use mesh analysis to set up two equations of the following form:

$$\begin{aligned} eI_1 - 2I_2 &= f \\ -2I_1 + gI_2 &= h \end{aligned}$$

Vraag 3.2 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} eI_1 - 2I_2 &= f \\ -2I_1 + gI_2 &= h \end{aligned}$$



Loop 1: $+2 + 5I_1 + 2(I_1 - I_2) + 2 = 0$

$$7I_1 - 2I_2 = -4$$

$$e = 7 \checkmark \quad f = -4 \checkmark$$

Loop 2: $-2 + 2(I_2 - I_1) + 3(I_2 - I_3) + 5 + 10I_2 = 0$

$$-2I_1 + 15I_2 - 3I_3 = -3$$

But $I_3 = -2A$

$$-2I_1 + 15I_2 - 3(-2) = -3$$

$$-2I_1 + 15I_2 = -9$$

$$g = 15 \checkmark \quad h = -9 \checkmark$$

Question 3.3 [4]

Use mesh analysis to set up two equations of the following form:

$$I_1 + mI_2 = n$$

$$pI_1 + qI_2 = \text{~~4~~ 4}$$

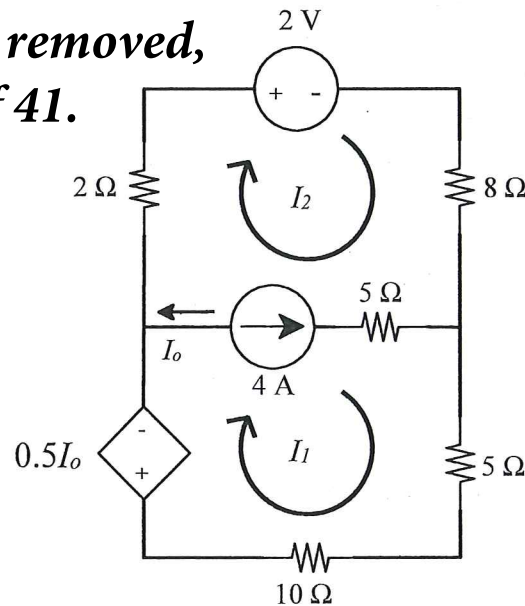
Question 3.3 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$I_1 + mI_2 = n$$

$$pI_1 + qI_2 = \text{~~4~~ 4}$$

**This question was removed,
so the test is out of 41.**



KVL Outside Loop:

$$+ 0.5 I_o + 2 I_2 + 2 + 8 I_2 + 5 I_1 + 10 I_1 = 0$$

$$0.5 I_o + 15 I_1 + 10 I_2 + 2 = 0$$

$$\text{But } I_o = -4A$$

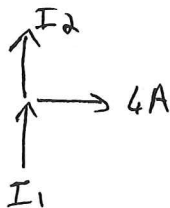
$$0.5 (-4) + 15 I_1 + 10 I_2 + 2 = 0$$

$$15 I_1 + 10 I_2 = 0$$

$$I_1 + \frac{10}{15} I_2 = 0$$

$$m = 0.67 \checkmark \quad n = 0 \checkmark$$

Supermesh:



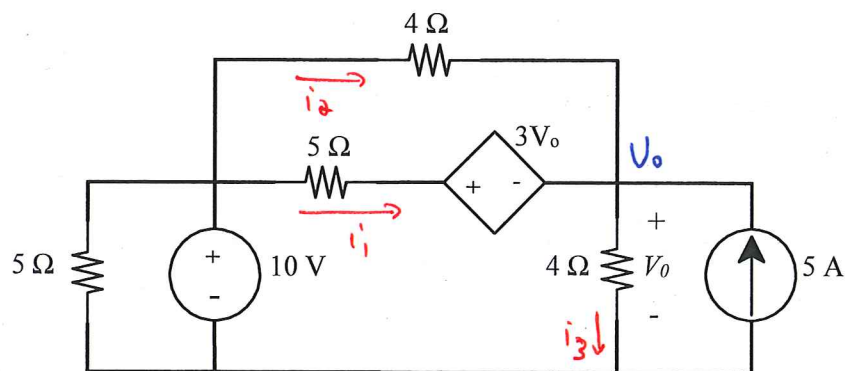
$$\text{KCL: } I_1 = I_2 + 4$$

$$I_1 - I_2 = 4$$

$$p = 1 \checkmark \quad q = -1 \checkmark$$

Question 3.4 [2]
Solve for V_o .

Vraag 3.4 [2]
Los op vir V_o .



At node V_o : $i_1 + i_2 + 5 = i_3$

$$\frac{10 - 3V_o - V_o}{5} + \frac{10 - V_o}{4} + 5 = \frac{V_o}{4}$$

(CCD: 20)

$$40 - 16V_o + 50 - 5V_o + 100 = 5V_o$$

$$190 = 26V_o$$

$$\underline{7.31V} = V_o$$

Question 3.5 [2]Determine α , if $v_2 = -5$ V and $v_3 = 0.5$ V.

$$\begin{bmatrix} 2 & 0 & 5 \\ 0 & 1 & 2\alpha \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$$

Question 3.5 [2]Bepaal α , as $v_2 = -5$ V en $v_3 = 0.5$ V.

$$\begin{bmatrix} 2 & 0 & 5 \\ 0 & 1 & 2\alpha \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$$

$$v_2 = \frac{\Delta_2}{\Delta}$$

$$\Delta = \begin{vmatrix} 2 & 0 & 5 \\ 0 & 1 & 2\alpha \\ 2 & 0 & 3 \end{vmatrix}$$

$$2 \cdot 1 \cdot 3 + 0 + 0 - 5 \cdot 1 \cdot 2 - 0 - 0$$

$$= 6 - 10$$

$$\Delta = -4$$

$$\Delta_2 = \begin{vmatrix} 2 & 4 & 5 \\ 0 & 0 & 2\alpha \\ 2 & 3 & 3 \end{vmatrix}$$

$$0 + 0 + 2 \cdot 4 \cdot 2\alpha - 0 - 2\alpha \cdot 3 \cdot 2 - 0$$

$$= 16\alpha - 12\alpha$$

$$\Delta_2 = +4\alpha$$

$$v_2 = \frac{\Delta_2}{\Delta}$$

$$-5 = \frac{4\alpha}{-4}$$

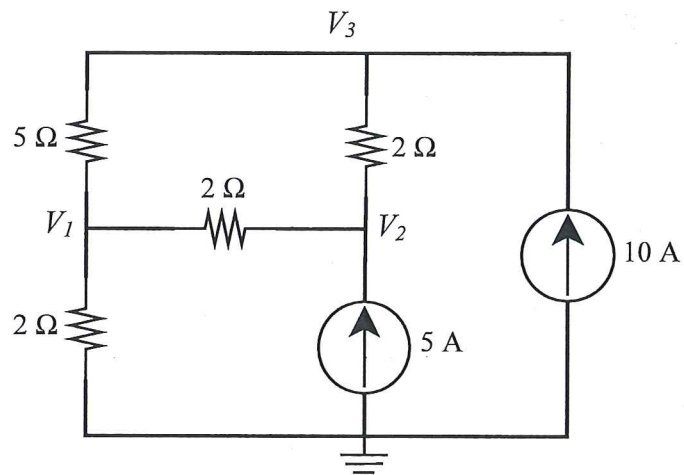
$$\alpha = 5 \checkmark \checkmark$$

Question 3.6 [3] [2]

Use inspection to set up a matrix equation in terms of the node voltages.

Vraag 3.6 [3] [2]

Stel 'n matriksvergelyking dmv inspeksie op in terme van die node-spannings.



$$\begin{bmatrix} 1.2 & -0.5 & -0.2 \\ -0.5 & 1 & -0.5 \\ -0.2 & -0.5 & 0.7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 10 \end{bmatrix}$$

$$G_{11} = \frac{1}{5} + \frac{1}{2} + \frac{1}{2} = 1.2$$

$$G_{22} = \frac{1}{2} + \frac{1}{2} = 1$$

$$G_{33} = \frac{1}{5} + \frac{1}{2} = 0.7$$

$$G_{12} = G_{21} = -\frac{1}{2} = -0.5$$

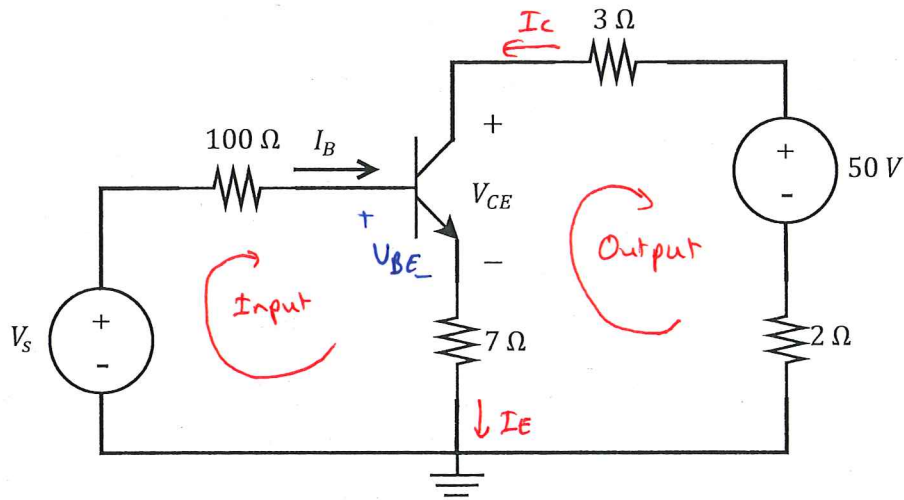
$$G_{23} = G_{32} = -\frac{1}{2} = -0.5$$

Question 3.7 [4]Given: $\beta = 100$, $I_E = 3 \text{ A}$ and $V_{BE} = 0.7 \text{ V}$.

- a) Determine V_S .
b) Determine V_{CE} .

Vraag 3.7 [4]Gegee: $\beta = 100$, $I_E = 3 \text{ A}$ en $V_{BE} = 0.7 \text{ V}$

- a) Bepaal V_S .
b) Bepaal V_{CE} .



a) Input Loop: $-V_S + 100 I_B + V_{BE} + 7 I_E = 0$

$$I_E = (1 + \beta) I_B \rightarrow \frac{I_E}{1 + \beta} = I_B$$

$$-V_S + \frac{100}{101} I_E + 0.7 + 7 I_E = 0$$

$$\frac{100}{101} (3) + 0.7 + 7(3) = V_S$$

$$V_S = \underline{24.67 \text{ V}}$$

b) Output loop:

$$-7 I_E - V_{CE} - 3 I_C + 50 - 2 I_C = 0$$

$$I_C = \beta I_B \rightarrow I_C = \frac{\beta}{1 + \beta} I_E = \frac{100}{101} I_E$$

$$-7 I_E - V_{CE} - 3 \left(\frac{100}{101} \right) I_E + 50 - 2 \left(\frac{100}{101} \right) I_E = 0$$

$$V_{CE} = +50 - 5 \left(\frac{100}{101} \right) (3) - 7(3)$$

$$= \underline{14.15 \text{ V}}$$

EBN122 SEMESTER TOETS/TEST 1 (25 AUG 2016)

ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans	English (Eng 3-1)	English (Eng 3-6)				

Q1		Q2		Q3		Tot:	/45 41
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VRAAG 1 – QUESTION 1 (8)

1.1	$R = 350.4 \checkmark \checkmark$
1.2	$\cancel{R} = J = -1.25 J \checkmark$
1.3	$P_1 = -63W \checkmark \checkmark$ $P_2 = 10W \checkmark$

VRAAG 2 – QUESTION 2 (16)

2.1	$P_{(1A)} = -7.5W \checkmark$	$P_{(7\Omega)} = 4.86W \checkmark$
2.2	$V_1 = 6V \checkmark$	$I_1 = 0.9A \checkmark$
2.3	$V_x = 1V \checkmark$	$V_y = -2.5V \checkmark$
2.4	$R_{ab} = 6.4\Omega \checkmark$	
2.5	$R_{cd} = 8.21\Omega \checkmark$	

VRAAG 3 – QUESTION 3 (22)

3.1	$a = -1 \checkmark$ -3	$b = 16 \checkmark$ 22	$c = -3 \checkmark$ -1	$d = 22 \checkmark$ 16
3.2	$e = 7 \checkmark$	$f = -4 \checkmark$	$g = 15 \checkmark$	$h = -9 \checkmark$
3.3	$m = 0.67 \checkmark$	$n = 0 \checkmark$	$p = 1 \checkmark$	$q = -1 \checkmark$
3.4	$V_0 = 7.31V \checkmark \checkmark$			
3.5	$\alpha = 5 \checkmark$			
3.6	$\begin{bmatrix} 1.2 & -0.5 \checkmark & -0.2 \\ -0.5 & 1 & -0.5 \\ -0.2 & -0.5 & 0.7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \checkmark \\ 5 \\ 10 \end{bmatrix}$			
3.7	$a) V_S = 24.67V \checkmark \checkmark$	$b) V_{CE} = 14.15V \checkmark \checkmark$		